

OpenCL+FPGA = System Acceleration!

Benefits of using the Open Computing Language (OpenCL) standard on FPGAs are higher performance, lower power consumption and faster time-to-market. How does a standard parallel programming language, coupled with the parallel performance capability of FPGAs, give such a powerful solution for system acceleration? Read on to know

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Initially developed by Apple and implemented by Intel, AMD, Nvidia, ARM and IBM to name a few, OpenCL is an industry standard managed and maintained by the Khronos Group. It is the first open, royalty-free, unified programming model for accelerating algorithms on heterogeneous systems.

The single-core CPUs that were initially used for computation, were soon replaced with multiple-core CPUs for higher performance. Today, there is a shift to comparatively smaller multi-core systems. Their small size is achieved by placing a general-purpose graphical processing unit (GPU), digital signal processor (DSP) or field-programmable gate array (FPGA) chips next to a CPU. However, engineers face the challenge of accelerating their programs across all of these different types of hardware. To tackle this challenge, they are looking for a single open standard programming model to bridge the hardware gap—and OpenCL could be the answer.

What is OpenCL?

For a software engineer OpenCL is a programming model, whereas for a system architect it is a methodology. OpenCL allows the use of a 'C'-based language for developing code across different platforms including CPUs, GPUs, DSPs and FPGAs. By utilising the standard 'C' language (C99) and OpenCL C (standard 'C' with some extensions), it can provide the solution for a single unified design flow in a different manner by using the open standard extension of 'C' language.

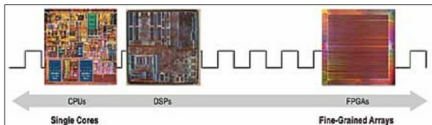


Fig. 1: Early spectrum of programmable technologies

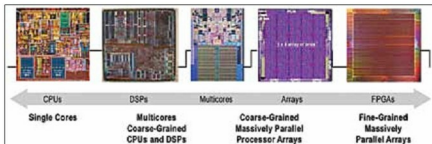


Fig. 2: Recent trend of programmable and parallel technologies

Design challenges to overcome while using FPGAs

1. Generally, programmers have difficulty using FPGAs as massive multi-core devices to accelerate parallel applications.
2. A programming model is needed that allows the designer to think about the FPGA as a configurable multi-core device.
3. Today, the FPGA's programming model revolves around RTL (VHDL/Verilog)—state machines, datapaths, arbitration, buffering, etc.
4. It is more difficult to develop an OpenCL program for an FPGA than more natural hardware platforms such as CPUs and GPUs.

Combining the OpenCL programming model with FPGA architecture provides a powerful solution for system acceleration. OpenCL is a standard multi-core programming model that can be used to provide a higher-level layer of abstraction for FPGAs.

An FPGA is basically a silicon chip that has a re-programmable digital circuitry. It can have thousands of configurable logic blocks connected with re-configurable wiring circuitry such that

these logic blocks can be configured in any manner for the required application. Certain sections of the FPGA can also be dedicated to a particular task, or a completely different task can run as a true parallel operation configured in a wholly different part of the chip.

What makes it tick?

The OpenCL host program is a pure software routine written in standard 'C' or 'C++' language that runs on